What is the effect of exposure to different concentrations (0, 2, 4 , 6, 8, 10%) of water soluble extracts within *Taraxacum officinale* leaves on the proliferation of *E. coli.* on petri dishes after 24 hours of growth at 37 °C as measured by the zone of inhibition diameter (mm) of the bacteria?

Background

A common type of Traditional Chinese Medicine (TCM) is herbal teas made via decoction, the process of boiling plant materials in water for a set amount of time before filtering out non-dissolved materials to obtain a water-soluble plant extracts solution (Ali *et al.* 2021). In TCM, the leaves of *Taraxacum officinale* (dandelion) are often used in herbal teas to treat illnesses such as bacterial infections (He *et al.* 2011). While multiple scientific studies have shown that many bioactive compounds in the tissue of the plant have antibacterial properties, the efficacy of the use of *T. officinale* in TCM herbal teas in combating bacterial infection is relatively unknown (Diaz *et al.* 2018). When I am sick, my mother often gives me TCM herbal teas as treatment, so I want to investigate the effects of dandelion tea in fighting against bacterial infections. Since decoction is used to made TCM herbal tea, this paper will focus on the effects of water-soluble extracts inside *T. officinale.*

Escherichia coli is a bacterium which grows in the lower intestine of warm-blooded animals. In humans the bacterium can cause many diarrheal illnesses (Skoog 2011). As a Gram-negative bacterium, *E. coli* has two lipid membranes whose fluidities are regulate through lipid homeostasis. The regulation of these membranes is especially crucial for cell division. Thus, any extrinsic changes to membrane fluidity will inhibit *E. coli* proliferation (Almaguer *et al.* 2022). Furthermore, membrane fluidity also modulates membrane receptors and enzymes. These protein structures are crucial for performing tasks necessary to *E. coli* survival such as the taking in nutrients and getting rid of metabolic waste. This demonstrates how changes in membrane fluidity can also indirectly inhibit bacterial proliferation (He *et al.* 2013).

A 2021 study exposing *E. coli* to different concentration (0% to 300%) of water-soluble compounds from dandelion leaf powder reveals a negative correlation between concentration of the compounds and *E. coli* proliferation (Li *et al*. 2021). This is likely due to a high concentration of the water-soluble flavonoids (natural compounds made by plants in response to microbial infections) luteolin and quercetin inside of the leaves of *T. officinale* (Olas 2022). According to a 2013 study on the impacts of flavonoids on *E. coli* membrane: both luteolin and quercetin decreases membrane fluidity in the bacterium. This is because they both possess hydrophobic prenyl chains which interact with the polar region of the membrane. The interaction increases ordering and the dynamics of membrane interior, causing the membrane to become more rigid. (He *et al*).

The inhibitory effects of certain substances on the proliferation of *E. coli* can be measured with a zone of inhibition test. With this method, filter paper disks laced with a known concentration of antimicrobial compound is place on an agar dish which has been inoculated with strains from a specific species of bacteria before the placing of the disks. The antibiotics will diffuse into the surrounding agar. Bacteria growth occurs in the presence of antibiotics when the bacteria reach a critical mass and can overpower the inhibitory effect of the antibiotic present. The critical mass point can be identified by the sharply

marginalized circles which surround the disks. These circles are known as zone of inhibition. The larger the diameter of the zone, the more inhibitory effect a substance had on bacterial growth (Hudzick 2009).

Hypothesis

As the concentration of water-soluble *T. officinale* leaves extracts increases, the zone of inhibition diameter of the *E. coli* also increases. This is because the extracts luteolin and quercetin are both water soluble and found in *T. officinale.* They react with the membranes of *E. coli* causing them to rigidify. The decrease in membrane fluidity impedes the bacterium's ability to perform cell division and prevents membrane proteins from performing functions necessary for *E. coli* survival. Both phenomena inhibit *E. coli* proliferation which leads to smaller the zones of inhibition.

Methodology

This IA aims to mimic the effects of TCM *T. officinale* herbal tea on bacterial infection. Therefore, it employs the traditional decoction method of water-soluble compound extraction: bringing a mixture (placed in it a 1000ml beaker) of 1-part dried dandelion leave (20 g) to 10-part water (200 g, distilled) to boil (at a heat of 10 on a Corning hot plate), leaving it to simmer for 20 minutes at a median heat (6 on the hot plate), and filtering out the insoluble part of the leave with a cheese cloth.

A preliminary experiment was conducted to estimate the concentration of water-soluble extract usually present in TCM by first measuring the mass of the 1000 ml beaker then creating an herbal tea with the method described above. After that the final mass of the solution is calculated by subtracting the mass of the solution in the beaker (measured with a balance) subtracted by the mass of the beaker. Then, the insoluble portion of the *T. officinale* was dried in an oven at 50 °C for 5 hours before being weighed with a balance. The mass of the dandelion extract left inside the herbal tea is calculated by subtracting the mass of the insoluble part of the leaves by 20g (initial weight). This number is then divided by the final mass of the herbal tea to find concentration (in %). These steps were repeated three times to reveal that there was about a 6% concentration of water-soluble compounds inside TCM herbal tea. Thus, in the final experiment, *E. coli* will be exposed to 0 (control group exposed to only distilled waper), 2,4,6,8,10 % (IV) of water-soluble *T. officinale* compounds to observe their effects when present in concentrations around the 6% mark. Meanwhile, the dependent variable will be the zone of inhibition of the *E. coli* 's diameter after 24 hours of growth at 37 °C. 24 hours was set as the time for incubation as the time frame gives *E. coli* enough time to proliferate while not being enough time for them to overcome the antibacterial effects of the herbal time and grow on top of the zone of inhibition. 37 °C was chosen as the incubation temperature as it mimics the temperature in the human body and therefore provides insight into whether TCM *T. officinale* herbal tea is effective in treating human diseases. DV will be measured with a larkpad ruler in mm.

<u>The day before:</u>

1. Prepare LB broth containing sodium chloride, peptone, and yeast

2. Prepare 6 LB 64 cm^2 petri dishes (on the bottom of the dish draw 3 lines across the dish to divide it into 6 equal parts using a permanent marker)
3. Prepare a starter plate with E. coli

<u>*T. officinale* herbal tea solution preparation:</u>

1. Used the tradition decoction method described above to prepare a tea solution.
2. Find out the mass of the dandelion extract left inside the herbal tea
3. Times the number by 10 to find the final volume (ml) the solution needs to be at to have 10% concentration of dandelion extract
4. Boil the solution on the hot plate until archiving the final volume by tracking it with the marks for volume on the beaker the solution is in.
5. Weigh the solution to check if it's final mass in g (subtract mass of beaker from total mass) is the same as the final volume in ml. If the mass is larger than the volume then repeat step 4 and 5 until archiving the right mass. If there the mass is smaller than the volume gradually add distilled water with a pipette until archiving the right mass.
6. Pour 10 ml of 10% solution into a test tube. Label test tube 10%.
7. Use the measurements below to dilute the remaining 10% solution into 10 ml of tea solution with 2,4,6,8 % concentration pour each solution into a test tube use a balance to measure the liquids needed. Label test tube.

Concentration (%)	Amount of 10 % solution needed (g)	Amount of distilled water needed (g)
2	2	8
4	4	6
6	6	4
8	8	2

8. Pour 10 ml of distilled water into a test tube. Label test tube.

<u>Plating E. coli and filter disks:</u>

1. Put on gloves and eye protection
2. Using a p1000 micropipette, collect 1200 µL of LB broth place it in a small sealable container
3. Using an inoculation loop, collect a small amount of *E. coli* bacteria and place it in the container
4. Seal and shake the container to mix the two components
5. Using a p1000 micropipette collect 200 µL of the mixture from the container and place it into an unused petri dish.
6. Using a glass spreader spread the bacteria evenly across the dish.
7. Turn the dish 90° clockwise
8. Repeat step 6 and 7 three more times
9. Repeat step 5 to 8 five more times.
10. Soak 6 filter disks into each of the test tubes prepared earlier
11. Label a plate 0% then place the disks in the 0% test tube on to the 6 sections drawn on the dish

12. Repeat step 11 for the disk in the other test tubes
13. Place all disk into incubator set to 37°C with the cap-side down to prevent condensation from dripping on the plates.

24 hours later: Take out the plate and record the diameter of the zone of inhibition around each disk for all the dishes with a larkpad ruler in mm. Record data in a table

Control Variable	Why should it be controlled?	How to control it?
Moisture inside leave before decoction	The more moisture in the leave at the beginning the heavier they are. This means when measuring 20 g of the plant for decoction, less leaves will be needed to get to this mass. Thus, less leaves and therefore less of water-soluble compound goes into the tea. This will affect the tea's ability to inhibit *E. coli* proliferation.	Dry the leaves in an oven for 50 °C for 5 hours before use.
Water for dandelion tea	There can be minerals and microorganisms present in water which would interfere with *E. coli* growth.	Distill water will be used
Temperature of the tea the paper disks are dipped in	If the herbal tea solution is freshly made it would be close to boiling therefore the disks that are dipped in it will also be at a temperature that is close to boiling. This temperature can kill off the *E. coli* in the petri dish thus leading to a larger zone of inhibition.	Allow the solution to cool to room temperature before use.
Size of filter paper disk	A larger paper disk leads to a larger zone of inhibition because to get to a certain diameter, herbal tea traveling from the edge of a larger disk goes over less distance to reach its destination compared to a smaller disk.	Disks with diameter of 6mm are used.
Temperature in incubator	A rise in temperature speeds up *E. coli* growth. This is because metabolic processes for growth speed up due to increasing collision rate between bacterial enzyme active sites and substrates. Also, when temperature is too high bacterial enzymes denature causing *E. coli* growth to halt.	*E. coli* will be kept at 37 °C
Time of incubation	Increase incubation time means *E. coli* would have longer time for growth. This increases the proliferation of *E. coli* and decreasing diameter of zone of inhibition.	The *E. coli* will be left in the incubator for 24 hours.

Safety Issues:
Direct contact with *E. coli* can cause damage to one's eyes and skin, and it can lead to stomach problems if ingested. Therefore, eye protections and gloves should be worn throughout the lab. After lab, all equipment which came in contact with the bacteria needs to be sanitized with ethanol and hands should be washed (Sigma Aldrich 2022).

Direct contact with ethanol can cause irritation to one's eyes and skin, so eye protections and rubber gloves should be worn throughout the lab. It is also a fire hazard and therefore needs to be kept away from sources of ignition (Fscimage).

Environmental Issues:
Nonreusable rubber gloves will be used throughout experiment. To reduce environmental impact, they should be recycled or disposed of in accordance to local Environmental disposal regulations (aureliagloves).

Ethical Issues: N/A

Raw Data:

Figure 1: Raw Data Table Showing the Extent Concentration of Water-Soluble Dandelion Extract (%) Impacts the Diameter of Zone of Inhibition (mm) of *E. Coli* after 24 Hours of Incubation at 37 °C

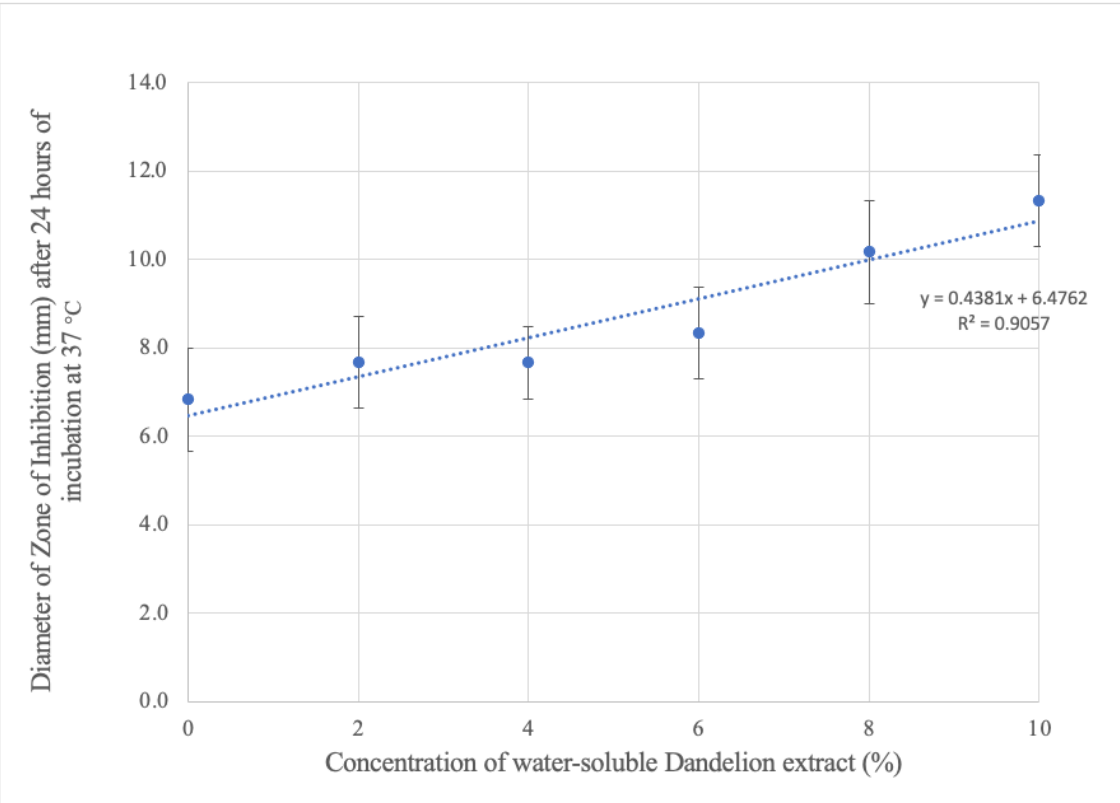
Concentration of water-soluble Dandelion extract (%)	**Diameter of Zone of Inhibition (mm)**					
	Trials (Petri Dishes)					
	0	1	2	3	4	5
0 (control)	8	7	5	6	8	7
2	7	8	8	6	9	8
4	8	9	7	7	8	7
6	8	8	7	10	8	9
8	9	10	9	11	10	12
10	13	11	11	10	11	12

Processed Data:

Figure 2: Processed Data Table Showing the Extent Concentration of Water-Soluble Dandelion Extract (%) Impacts the Diameter of Zone of Inhibition (mm) of *E. Coli* after 24 Hours of Incubation at 37 °C

Concentration (%)	**Average Diameter (mm)**	**Standard Deviation**
0 (control)	6.8	1.2
2	7.7	1.0
4	7.7	0.8
6	8.3	1.0
8	10.2	1.2
10	11.3	1.0

Figure 3: Graphical Representation of the Extent Concentration of Water-Soluble Dandelion Extract (%) Impacts the Diameter of Zone of Inhibition (mm) after 24 Hours of Incubation at 37 °C *Error bar is of STDV*



Sample Calculation:

$$STDV = \sqrt{\frac{\sum_1^N (x_i - \mu)^2}{N}}$$

Measure of Uncertainty:

Material Measured	Inaccuracy in Measurement	Impact on data
Mass of solution	Measured with balance with an inaccuracy of $\pm 0.01g$ (scalesplus).	Affects calculation for concentration of water-soluble compounds which impacts diameter of zone of inhibition.
Volume of *E. coli*	Measured with micropipette with an inaccuracy of ± 0.05 µL (labequipment)	Affects the amount of *E. colin* on a petri dish which impacts diameter of zone of inhibition.

Volume of LB solution	Measured with micropipette with an inaccuracy of ± 0.05 µL (labequipment)	Affects amount of nutrient *E. coli* gets which affects diameter of zone of inhibition.
Diameter of zone of inhibition	Larkpad ruler's inaccuracy not published	Affects data collect on diameter of zone of inhibition.

Interpretation of data:

Trend of data:

According to Figure 3, as the concentration of *Taraxacum officinale* leaf's water-soluble extracts increases, the diameter of the zone of inhibition of *E. coli* increases in a linear fashion at a predicted interval of about 0.4381 mm with every 1% increase in the concentration of the extracts. The R squared value is 0.9057 meaning about 91% of variation in data is accounted for by the trend line. This demonstrates a relatively strong correlation between the two variables. Figure 2 shows that changes in concentration from 0% to 4% led to less increase in average zone of inhibition diameter compared to changes in concentration from 6% to 10 %. From 2% to 4%, there was the least amount of change in the zone of inhibition diameters (an average of 0 mm) while from 6 to 8 %, changes in concentration led to the greatest change in the diameter (an average increase of 1.9 mm according to Figure 2). From 0% to 10 % concentration of *T. officinale* leaves water-soluble extracts, there was an average increase of 4.5 mm in zone of inhibition diameter.

Variation in data:

According to Figure 2, the experimental groups all have relatively large standard deviations with the group with 4% water-soluble extracts having the smallest standard deviation of 0.8 and the groups 0% and 8% having the largest standard deviation of 1.2. This data indicates that the average zone of inhibition diameter of E. coli is not a good predicator for what the actual diameter for zone of inhibition when *E. coli* is exposed to certain concentration of water-soluble *Taraxacum officinale* leaf extract is. Furthermore, according to Figure 3, there are overlaps between the error bars of all pair of experimental groups that are 2% apart (ie. between 0% and 2%).

The fnull hypothesis is that there this is no significant difference between experimental groups: 0% and 6%. An unpaired T-test was performed between the control group (0%) and 6% water-soluble extracts within *Taraxacum officinale* leaves. The p value was 0.0403. This indicates that the difference between the data set is statistically significant, because the p value is below 0.05. The null hypothesis is proven to be wrong.

Conclusion

According to the data collected, the hypothesis that in the range of 0 to 10%, as the concentration of water-soluble *Taraxacum officinale* leaf extract increases, the diameter of the zone of inhibition for *E. coli* will increase too is somewhat supported. While the trend of the data as shown in the Figure 3 does demonstrate a relatively strong positive correlation between the two variables, the high variance in the data makes it unclear if the trend is statistically significant. When examining the error bars, the overlap

between error bars of all pairs of experimental groups that are 2% apart suggests the effects a 2 % increase in concentration of water-soluble extract has on the diameter of the zone of inhibition is statistically insignificant. The lack of clear data may be caused by two things. Firstly, the concentration of water-soluble *Taraxacum officinale* leaf extract may be too small to have a significant amount of luteolin and quercetin which is necessary to inhibit *E. coli* bacterial proliferation. This is supported by the results from the 2021 study done on dandelion water-soluble extracts' antibacterial abilities. According to their data only after there are 100% or higher concentrations of water-soluble extracts did the antibacterial effect of the extract become absolutely apparent (Li *et al*. 2021). Secondly, the process used to extract water-soluble compounds from *T. officinale* may have led to the decomposition of luteolin. While quercetin is shown to have high thermal stability (Elhamirad and Zamanipoor, 2012). A 2017 study shows that, exposing luteolin to 110 °C heat causes significant decomposition of the chemical. Specifically, when left under the 110 °C heat for 20 minutes, conditions similar to what luteolin experienced during the extraction technique (decoction) used in the lab (left in boiling water for 20 minutes), 40 % of luteolin decomposed. It can therefore be inferred that decoction likely caused a significant decrease in luteolin thus weaking the inhibitory effect of *T. officinale* water-soluble extracts on *E. coli* proliferation leading to smaller increases in diameter of zone of inhibition as concentration of water-soluble extract increases than what was hypothesized.

Although the TCM technique of decoction likely decreased the concentration of luteolin extracted and 2% changes in concentration of water-soluble extracts are not enough to impact *E. coli* growth inhibition, the result of this experiment does provide evidence that dandelion herbal teas have some anti-bacterial capabilities. As shown by the p value of an unpair T-test between the control group (0%) and 6%, the chances that the increase in diameter (1.5 mm) of the zone of inhibition is due to chance rather than increase in concentration is less than 5 %. According to preliminary research, there are about 6% of water-soluble *Taraxacum officinale* leaf extract inside a CTM herbal tea. It can be inferred from the data, the amount of luteolin and quercetin inside the 6% solution is enough to halt E. coli proliferation. According to research, when ingested by human, flavonoids are shown to be able to fight against infectious diseases (Kumar and Pandey, 2013). This means that drinking an herbal tea with 6% of water-soluble *Taraxacum officinale* leaf extract inside may be able to help with *E. coli* infection in humans.

Evaluation

<u>Strength:</u>

I performed preliminary research to find the average concentration of water-soluble compound in TCM herbal tea. I then used the result of my research to accurately stimulate the effect of TCM herbal tea on *E coli*. by testing the inhibitory effect of water-soluble compounds in a concentration range that is likely to be found in the herbal tea. I also used decoction for extraction of water-soluble compound from *T. officinale.* This allows me to stimulate the condition of TCM herbal tea. I also stimulate condition inside a human body by incubating the bacteria at a normal body temperature of 37 °C. All these choices help me reach my goal for this IA of investigating the efficacy of *Taraxacum officinale* TCM herbal tea in

treating *E. coli* infections in humans. Another strength of my research is the use of zone of inhibition test to test *E. coli* proliferation. This method is simple and therefore can be easily replicated.

Limitation	Its impact	Improvement for next time
Time filter pater disk spent soaked in herbal tea solution was not control	The more time the paper spends in the solution the more of the solution it can soak up. When paper discs with more solution are placed on the petri dish, the bacteria would be exposed to more water-soluble *T. officinale* extract which impacts the diameter of the zone of inhibition.	Measure the length of time the paper is soaked in the solution using a stopwatch to make sure each disk only spends 30 seconds inside the solution before being taken out.
Only one batch of *T. officinale* herbal tea was made and used as IV for	If that batch of tea was made incorrectly or was contaminate, the reliability of the entire study would be compromised.	Use a different batch of tea for each trial to minimalize the impact of a faulty batch of tea on the data collected.
Use of Larkpad rulers to measure zone of inhibition	The smallest unit of measurement on the ruler is mm. Millimetre isn't exact enough as the dimeter of the zone of inhibition is quite small. This introduces uncertainty to the data collected which makes the results of the investigation less reliable.	Use the software Imagej which can scan photographs taken of petri dishes and digitally measures the diameter of the zone of inhibition. The software has a small measure of uncertainty of 0.005 mm.

<u>Extension:</u>

A potential extension would an investigation on how different concentrations of *T. officinale* water-soluble compounds (0,2,4,6,8,10%) extracted using a different method of extraction impact *E. coli* growth. Comparing the results of this potential study with the current study would reveal whether or not decoction decreased the concentration of luteolin in a *T. officinale* extract as hypothesized in the conclusion. Another extension would be investigating how a larger range of concentration of *T. officinale* water-soluble compounds (0, 10, 20, 30,40, 50,60,90,100%) extracted using decoction would impact *E. coli* growth. As TCM herbal tea is made via decoction, this lab would help figure out the concentration water-soluble compounds needed to made the most effective TCM herbal tea for curing *E. coli* infections

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